

## UJUNG PANDANG HASANUDDIN, Indonesia

WMO: 971800

Lat: 5.0711S

Lon: 119.5522E

Elev: 14

StdP: 101.16

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 20.1       | 21.0 | 13.2                          | 9.5 | 28.0 | 15.0 | 10.7 | 27.6 | 7.8                   | 28.0 | 6.7 | 28.2 | 1.2                   | 180  | 0.310 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 10            | 10.5                   | 34.6            | 23.9 | 34.0 | 24.3 | 33.2 | 24.8 | 28.2                | 30.7 | 27.7 | 30.5 | 27.2 | 30.4 | 4.2                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 27.8                            | 23.8 | 29.2 | 27.1 | 22.8 | 28.8 | 26.5 | 22.1 | 28.5 | 90.9          | 30.9 | 88.4 | 30.9 | 86.4 | 30.6 | 30.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 6.5           | 5.7  | 5.0 | DB                         | 18.3 | 36.3               | 1.1 | 1.3                                                | 17.5 | 37.3       | 16.8 | 38.0       | 16.2 | 38.7       | 15.4 | 39.7 | (4) |
| (5)               |      |     | WB                         | 15.8 | 28.8               | 1.5 | 0.9                                                | 14.8 | 29.4       | 13.9 | 29.9       | 13.1 | 30.4       | 12.0 | 31.0 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 27.5        | 26.9  | 27.0  | 27.3  | 27.7  | 27.9  | 27.5  | 27.0  | 27.3  | 27.8  | 28.2  | 28.0  | 27.2  | (6)  |  |
|                                                                        | DBStd       | 1.04        | 1.09  | 1.03  | 0.94  | 0.86  | 0.83  | 0.89  | 0.93  | 1.00  | 0.95  | 0.84  | 0.91  | 1.14  | (7)  |  |
|                                                                        | HDD10.0     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (8)  |  |
|                                                                        | HDD18.3     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (9)  |  |
|                                                                        | CDD10.0     | 6379        | 523   | 476   | 537   | 531   | 556   | 525   | 527   | 536   | 533   | 563   | 539   | 533   | (10) |  |
|                                                                        | CDD18.3     | 3338        | 265   | 242   | 279   | 281   | 298   | 275   | 269   | 278   | 283   | 304   | 289   | 274   | (11) |  |
|                                                                        | CDH23.3     | 33461       | 2274  | 2118  | 2589  | 2738  | 3079  | 2712  | 2654  | 3063  | 3386  | 3484  | 2923  | 2439  | (12) |  |
| (13)                                                                   | CDH26.7     | 13426       | 664   | 662   | 903   | 1019  | 1218  | 1065  | 1118  | 1439  | 1691  | 1682  | 1194  | 772   | (13) |  |
| (14)                                                                   | Wind        | WSAvg       | 1.9   | 2.0   | 2.0   | 1.9   | 1.7   | 1.6   | 1.8   | 2.1   | 2.3   | 2.1   | 1.9   | 2.0   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 3460        | 766   | 588   | 431   | 274   | 166   | 98    | 60    | 24    | 41    | 117   | 277   | 619   | (15) |  |
|                                                                        | PrecMax     | 5195        | 1301  | 1181  | 757   | 565   | 333   | 221   | 311   | 115   | 347   | 413   | 686   | 1059  | (16) |  |
|                                                                        | PrecMin     | 2580        | 156   | 99    | 209   | 119   | 59    | 21    | 6     | 8     | 12    | 1     | 70    | 231   | (17) |  |
|                                                                        | PrecStd     | 703         | 234   | 242   | 166   | 102   | 71    | 63    | 72    | 25    | 64    | 105   | 146   | 219   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 32.8  | 32.4  | 33.1  | 33.5  | 33.8  | 33.8  | 33.7  | 34.7  | 35.5  | 36.1  | 34.5  | 33.1  | (19) |  |
|                                                                        |             | MCWB        | 27.2  | 26.5  | 26.9  | 26.7  | 26.4  | 25.4  | 24.5  | 23.8  | 22.6  | 22.6  | 25.6  | 26.8  | (20) |  |
|                                                                        | 2%          | DB          | 31.7  | 31.8  | 32.2  | 32.8  | 33.0  | 32.9  | 32.9  | 33.8  | 34.4  | 34.6  | 33.5  | 32.1  | (21) |  |
|                                                                        |             | MCWB        | 26.5  | 26.4  | 26.5  | 26.6  | 26.3  | 25.3  | 24.5  | 23.6  | 23.4  | 23.8  | 25.8  | 26.6  | (22) |  |
|                                                                        | 5%          | DB          | 30.9  | 31.1  | 31.8  | 32.1  | 32.2  | 32.1  | 32.1  | 33.0  | 33.8  | 33.9  | 32.9  | 31.3  | (23) |  |
|                                                                        |             | MCWB        | 26.4  | 26.3  | 26.4  | 26.3  | 26.1  | 25.4  | 24.4  | 23.7  | 23.5  | 24.3  | 25.9  | 26.5  | (24) |  |
|                                                                        | 10%         | DB          | 30.0  | 30.2  | 30.9  | 31.2  | 31.8  | 31.2  | 31.3  | 32.2  | 33.0  | 33.0  | 32.0  | 30.7  | (25) |  |
|                                                                        |             | MCWB        | 26.0  | 26.1  | 26.2  | 26.1  | 26.0  | 25.3  | 24.3  | 23.6  | 23.6  | 24.7  | 26.0  | 26.3  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 28.4  | 28.2  | 28.3  | 28.4  | 28.3  | 27.7  | 26.8  | 26.4  | 27.2  | 27.8  | 28.6  | 28.3  | (27) |  |
|                                                                        |             | MCDB        | 30.6  | 30.1  | 30.9  | 30.7  | 31.0  | 30.3  | 30.6  | 30.9  | 30.6  | 30.8  | 31.3  | 30.6  | (28) |  |
|                                                                        | 2%          | WB          | 27.3  | 27.3  | 27.5  | 27.5  | 27.6  | 27.0  | 26.2  | 25.7  | 26.2  | 26.9  | 27.7  | 27.6  | (29) |  |
|                                                                        |             | MCDB        | 29.9  | 30.0  | 30.4  | 30.8  | 30.7  | 30.3  | 29.9  | 30.4  | 30.4  | 30.8  | 30.8  | 30.3  | (30) |  |
|                                                                        | 5%          | WB          | 26.8  | 26.8  | 26.9  | 27.0  | 27.1  | 26.5  | 25.8  | 25.2  | 25.6  | 26.4  | 27.0  | 27.0  | (31) |  |
|                                                                        |             | MCDB        | 29.4  | 29.6  | 30.1  | 30.5  | 30.4  | 29.7  | 29.5  | 30.2  | 30.2  | 30.5  | 30.6  | 30.1  | (32) |  |
|                                                                        | 10%         | WB          | 26.3  | 26.3  | 26.5  | 26.6  | 26.6  | 26.1  | 25.3  | 24.7  | 25.0  | 25.9  | 26.5  | 26.5  | (33) |  |
|                                                                        |             | MCDB        | 28.7  | 29.0  | 29.5  | 29.9  | 29.8  | 29.2  | 29.2  | 29.9  | 30.0  | 30.2  | 30.0  | 29.3  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 6.2   | 6.6   | 7.2   | 7.8   | 8.3   | 8.6   | 9.7   | 11.0  | 11.5  | 10.5  | 8.5   | 6.6   | (35) |  |
|                                                                        |             | MCDBR       | 7.5   | 7.8   | 8.3   | 8.6   | 9.0   | 9.6   | 10.7  | 11.8  | 12.6  | 12.1  | 9.8   | 7.9   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 3.4   | 3.6   | 3.7   | 3.8   | 3.8   | 3.9   | 4.3   | 4.8   | 5.3   | 5.2   | 4.4   | 3.6   | (37) |  |
|                                                                        |             | MCWBR       | 6.8   | 7.2   | 7.6   | 8.0   | 8.2   | 8.4   | 9.1   | 10.3  | 10.4  | 9.6   | 8.7   | 7.4   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.435 | 0.435 | 0.427 | 0.418 | 0.416 | 0.417 | 0.407 | 0.416 | 0.436 | 0.477 | 0.476 | 0.442 | (40) |  |
|                                                                        |             | taud        | 2.422 | 2.404 | 2.434 | 2.473 | 2.481 | 2.481 | 2.500 | 2.445 | 2.366 | 2.242 | 2.270 | 2.402 | (41) |  |
|                                                                        | Ebn at Noon |             | 904   | 902   | 898   | 882   | 856   | 841   | 857   | 871   | 877   | 855   | 860   | 894   | (42) |  |
|                                                                        |             | Edh at Noon | 123   | 126   | 121   | 112   | 106   | 104   | 103   | 113   | 127   | 146   | 142   | 125   | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 4.61  | 4.88  | 5.31  | 5.39  | 5.20  | 4.84  | 5.23  | 6.02  | 6.57  | 6.49  | 5.74  | 4.65  | (44) |  |
|                                                                        | RadStd      |             | 0.44  | 0.41  | 0.35  | 0.32  | 0.25  | 0.39  | 0.37  | 0.31  | 0.33  | 0.52  | 0.44  | 0.42  | (45) |  |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (2 neighbors) | +0.24 | N/A     | N/A    | +0.55   | +0.72 | +0.88 | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page